CODE DIGEST
All your tech in one place

sfi
CET

ACADEMIC SUPPORT CELL
An Initiative by SFI CET Unit

APP
DEVELOPMENT

PHASE 01

Programming Fundamentals

Technologies & Libraries
- Java
- Kotlin (Android)
- Dart (Flutter)

Familiarize Yourself With
- VS Code
- Android Studio
- IntelliJ IDEA
- Git and GitHub
- Terminal /Command Line

Build
- Calculator
- Number Guessing Game

Learn
- Variables and Data Types
- Operators
- Conditional Statements
- Loops
- Functions
- Arrays and Collections
- Strings and String Manipulation
- OOP Concepts
- Problem Solving
- Basic Algorithm and Logic

PHASE 02

Mobile App And UI Development

Technologies & Libraries
- Android SDK
- Flutter SDK
- XML
- Jetpack Compose
- Material UI

Familiarize Yourself With
- Material Design
- Figma
- Lottie
- Google Fonts
- Wireframing and Prototyping
- Emulator and Device Testing

Learn
- Mobile App Architecture
- UI Components and Widgets
- Layouts & Responsive UI
- Navigation
- Activity Lifecycle
- Forms and User Input
- Animations
- Themes and Styling
- Typography, Colors and Spacing

Build
- Login Screen
- Notes App UI
- Weather UI
- E-commerce UI

PHASE 03

State Management

Learn
- State
- Widget Lifecycle
- Data Flow
- Local State
- Global State
- State Updates

Technologies & Tools
- Provider
- Riverpod

Familiarize Yourself With
- MVVM
- Clean Architecture
- Bloc
- GetX

Build
- Expense Tracker
- Shopping Cart
- Quiz App

PHASE 04

Backend Integration

Familiarize Yourself With
- Postman
- Swagger
- Retrofit
- Express.js
- FastAPI
- HTTP Status Codes

Technologies & Tools
- Dio

Learn
- REST APIs
- JSON
- HTTP Requests
- API Parsing
- Error Handling

Build
- Weather App
- News App
- Movie App

PHASE 05

Data And Authentication

Learn
- CRUD and Data Persistence
- Local and Offline Storage
- SQL and Database Basics
- Authentication and Authorization
- JWT and OAuth
- User Sessions

Technologies
- SQLite
- Hive
- Shared Preferences
- Firebase Authentication
- Supabase Auth

Familiarize Yourself With
- ORM Concepts· Database Migrations
- Local Caching
- Google Login
- Email and OTP Verification
- Password Reset
- Two-Factor Authentication
- Role-Based Access Control

Build
- Offline Notes App
- Expense Tracker
- Secure Login App

PHASE 06

Advanced Features

Learn
- Push Notifications
- Maps and Location
- Camera and Image Capture
- GPS and Geolocation
- File Upload and Storage
- Payments

Build
- Food Delivery App
- Cab Booking App
- Social Media App

Technologies
- Firebase Cloud Messaging
- Google Maps API
- Razorpay
- Stripe

Familiarize Yourself With
- Permissions
- Background Services
- Image Compression
- Location Tracking

PHASE 07

Testing, Optimization And Deployment

Learn
- Testing and Debugging
- Performance and Memory Optimization
- APK and AAB Generation
- Versioning and App Updates
- Play Store Publishing
- CI/CD and Build Automation
- Release Management

Familiarize Yourself With
- Performance Metrics
- Crash Reporting and Logging
- Test Coverage and Mocking
- Production Debugging

Technologies
- Flutter Test
- Mockito
- DevTools
- Firebase Crashlytics
- Google Play Console
- GitHub Actions
- Codemagic

Build
- Test and Optimize Existing Apps
- Publish an App
- Release an App Update
- Build an App Portfolio